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Memory:

For the Award of the LESANSE Diploma in LTS

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Solve a problem in welding the Godet

Publicly Defended

Before the Jury Composed of:

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College year:
2022/2023

Thanks:

We first thank God who has given us the will to complete this work, which represents the culmination of our years of study.

*We would like to express our sincere thanks and deep gratitude to **Dr MEKROUD Abdeslem,** Lecturer at the University of Mentouri Brothers Constantine1, for his patience, availability, and especially his advice, which has contributed to fueling our thinking throughout this work.*

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Dedication:

We dedicate this modest work, which represents many years of study, and we hope it lives up to our efforts, to all our families.

Firstly, to our dear fathers and tender mothers who have always been present in every moment of our lives, and also to our sisters and brothers.

We also dedicate this project to all those who are very dear to us and who have provided invaluable help that we will never forget.

Idriss.

Dedication:

To all our families, this humble endeavor symbolizes years of dedication and study. We aspire for it to reflect the magnitude of our efforts.

Firstly, to our beloved fathers and nurturing mothers, whose unwavering presence has shaped every moment of our lives. We extend this dedication to our siblings as well.

Additionally, we dedicate this project to those dear to us, whose invaluable support will forever remain ingrained in our hearts.

Haroune.

Dedication:

We dedicate this modest work to all our families, symbolizing years of dedication and hard work in study. We hope this work reflects the magnitude of our efforts.

First and foremost, to our dear fathers and nurturing mothers who have always been present in every moment of our lives, as well as to our siblings.

We also dedicate this work to those who hold a special place in our hearts, and who have provided us with the necessary support and assistance that we will never forget.

Ramzi.

ملخص:

قطاع تصنيع آلات البناء نشط للغاية ومربح لكنه لازال يتأثر بالأزمات المالية تتطلع الشركة الى حل المشاكل التي تواجهها و تحسين منتجاتها لارضاء العملاء و الحصول على مكان خاص في السوق بعيدا عن هذه الازمة في هذا السياق اتيحت لنا الفرصة للمساعدة في الحل مشكلة متعلقة بالتلحيم لأحد الهياكل خلال فترة التربص التي استمرت لثلاث أشهر في هذه الشركة وقد ساعدنا في حل مشكلة التلحيم . تم توجيه الدراسة أولا نحو العرض التقديمي للشركة وبعض منتجات الشركة أما الجزء الثاني فيطرح المشكلة المتعلقة بالتلحيم بالتفصيل و المواد و الطرق المستخدمة لذلك و الجزء الأخير سندرس المشكلة معا ونعطي فرضيات قد توصلنا لحل جديد و أفضل يساعدنا على حل المشكلة بشكل نهائي اعتمادا على ما قد درسناه مسبقا ثم نختم هذه الرسالة بالمراجعة و الآراء.

Summary:

In a sector as active and profitable as construction machinery manufacturing, the lingering effects of financial crises persist. **ENMTP** (**N**ational **C**ompany of **M**aterials and **P**ublic **W**orks) Companies in this industry aim to confront challenges and enhance their products to meet customer expectations and secure a prominent position in the market, away from the ramifications of these crises. In this context, we had the opportunity during our three-month internship at this company to contribute to resolving a welding-related issue for a specific structure. Our study commenced with an overview of the company and some of its products. Subsequently, we delved into the welding problem in detail, exploring the materials and methods involved. In the final segment, we collectively analyzed the issue and proposed hypotheses for a novel and improved solution based on our previous studies. Finally, we concluded our report with a comprehensive review and feedback collection.

Résumé :

Dans un secteur aussi actif et rentable que la fabrication de machines de construction, les effets persistants des crises financières persistent. **ENMTP (E**ntreprise **N**ational des **M**atériels des **T**ravaux **P**ublique**)** Les entreprises de ce secteur cherchent à relever les défis et à améliorer leurs produits pour répondre aux attentes des clients et sécuriser une position de premier plan sur le marché, loin des répercussions de ces crises. Dans ce contexte, nous avons eu l'occasion, lors de notre stage de trois mois dans cette entreprise, de contribuer à résoudre un problème lié au soudage pour une structure spécifique. Notre étude a débuté par un aperçu de l'entreprise et de certains de ses produits. Par la suite, nous avons approfondi le problème de soudage en détail, en explorant les matériaux et les méthodes impliqués. Dans le dernier segment, nous avons analysé collectivement le problème et proposé des hypothèses pour une solution nouvelle et améliorée basée sur nos études précédentes. Enfin, nous avons conclu notre rapport par une revue complète et la collecte de feedbacks.

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General Introduction

This current work falls within the framework of presenting a thesis for the completion of studies in the professional license in **Welding Technology (LTS)**, in an industrial environment characterized by intense competitiveness, where the company finds itself today, more than ever, compelled to achieve the goals: quality, cost, and timelines. In order to maintain this balance, the company seeks to eliminate all defects and wastage in production, whether it be in production time, raw materials, weight, high vibrations, or noise; starting from the principle that every "problem" is an opportunity for improvement and should be addressed as such in its activities.

ENMTP is one of the leaders in manufacturing, specifically in the field of equipment, due to its focus on design, production, welding, and sales of excavation and compression machinery, and other general construction equipment [1].

The company is divided into several departments, including the **"SOFARE"** department for complex compressors and compressors in **Ain-Smara**, where we were assigned a research task.

The main objective of this work is to solve a recurring welding problem, based on the specific goal, to reach a final solution to the welding problem. In the first chapter, an overview of the National Company for Public Works Equipment "ENMTP" is given: after a reminder of the manufacturing processes. In the second chapter, the problem related to welding that the company frequently faces and all the tools and equipment used for it will be presented, along with a general explanation of the **backhoe loader** where this problem occurs continuously. The third chapter will address a comprehensive study of this problem and the development of a final solution, with a reminder of what we have learned in the basics of welding and the appropriate materials for it. In the final chapter, we will review all the previous work and provide our conclusions reached through our solution.

Chapter I:

Company presentation

I. COMPANY PRESENTATION

I.1. NATIONAL COMPANY OF PUBLIC WORKS EQUIPMENT (ENMTP)

"ENMTP," was established on January 1, 1980, following the restructuring of SONACOME and SN METAL. ENMTP was transformed into a Joint Stock Company in 1995. Its social capital amounts to 15,600,000,000 Algerian dinars, fully owned by the state and managed by the Mechanical Group. Its current activities include:

Design, development, production, marketing, and maintenance of a wide range of public works equipment related to earthmoving, lifting, handling, compacting, concrete preparation, and other materials, as well as equipment for compressed air and asphalt.

ENMTP manufactures equipment under licenses acquired from internationally renowned manufacturers such as:

- LIEBHERR (Germany) for excavators, cranes, bulldozers, and large loaders.
- O & K (Germany) for wheel loaders.
- INGERSOLL RAND CO. (USA) for compressors and compactors.
- POTAIN (France) for tower crane manufacturing.
- BRAUD & FAUCHEUX for concrete mixers.

ENMTP has also independently developed equipment such as graders, backhoe loaders, BTS presses, and various small products, using internal expertise [2].

I.2. ENMTP Branches

❖ CONSTANTINE:

- SOMATEL subsidiary.

- JV SOMATEL-LIEBHERR.

- SOFARE subsidiary.

- JVEUROPACTOR ALGERIA.

- Maintenance and Distribution Unit **Ain-Smara** Constantine.

❖ ALGER:

- SOMATEL subsidiary
- Maintenance and Distribution Unit Alger

❖ ANNABA:

- Maintenance and Distribution Unit Annaba

❖ BÉJAÏA:

- FAGECO subsidiary

❖ ORAN:

- Maintenance and Distribution Unit Oran

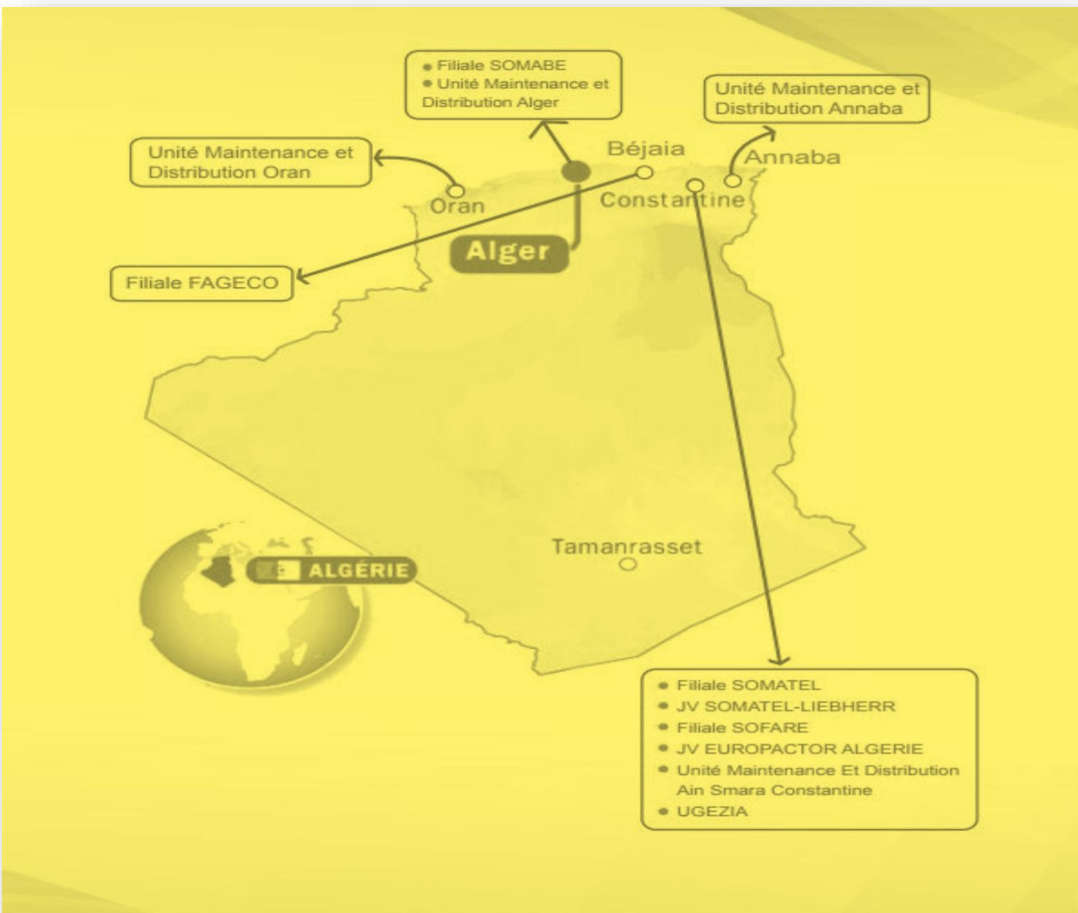

Fig. I.1. The branches of the ENMTP

I.3. ENMTP Subsidiaries

ENMTP has five (05) production subsidiaries (03 in Ain-Smara, 01 in Alger, and 01 in Bejaïa) and one (01) service subsidiary:

- **SOMATEL** (Production of Excavators, Loaders, Graders) subsidiary located in Ain-Smara Constantine.
- **SOMATEL-LIEBHERR:** A subsidiary of SOMATEL (Production of New Generation Earthmoving Equipment).
- **SOFAME** (Mechanical Manufacturing) subsidiary located in Ain-Smara Constantine.
- **SOFARE** (Production of Backhoe Loaders, Compressors, Compactors, and Concrete Pumps) subsidiary located in Ain-**Smara** Constantine.
- **EUROPACTOR ALGERIA:** A subsidiary of SOFARE (Production of New Generation Compaction Equipment).
- **FAGECO** (Production of Asphalt Distributors and Binder Spreaders) subsidiary located in Bejaïa.
- **SOMABE** (Production of Concrete Equipment: Concrete Mixers, Construction Dumpers, and Block Making Machines) subsidiary located in El-Harrach Alger.

Regarding our final project, we chose the SOFARE subsidiary due to the availability of significant activity volume compared to other subsidiaries of the ENMTP group.

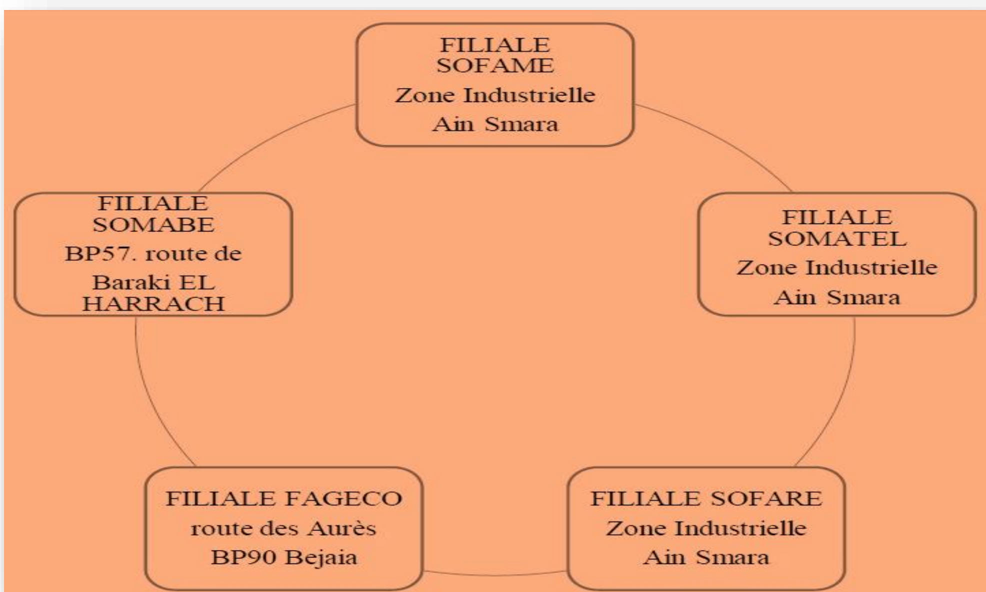

Fig. I.2. ENMTP subsidiaries.

I.4. Presentation of SOFARE

The Compact Compactors Compressors Complex in Ain-Smara, abbreviated as CCA, whose production began in 1985, was transformed into a Joint Stock Company (SPA) named SOFARE (Backhoe Loader Manufacturing Company) in April 2011. This subsidiary is 100% owned by the ENMTP Group.

SOFARE is located in the Industrial Zone of the ENMTP Group in Ain-Smara, Constantine province. It covers an area of nearly 221,545 square meters, of which 54,352 square meters are covered. It is 8 km from the Mohamed Boudiaf Airport in Constantine, 500 meters from the East-West Highway, and 11 km from the capital of the province [2].

I.5. Presentation of the CCA Complex

The compactors complex is located in the industrial zone of "AIN SMARA" on Route No. 05. Following agreements on February 18, 1977, between ENMTP and the American company (Ingersoll Rand), the complex was created.

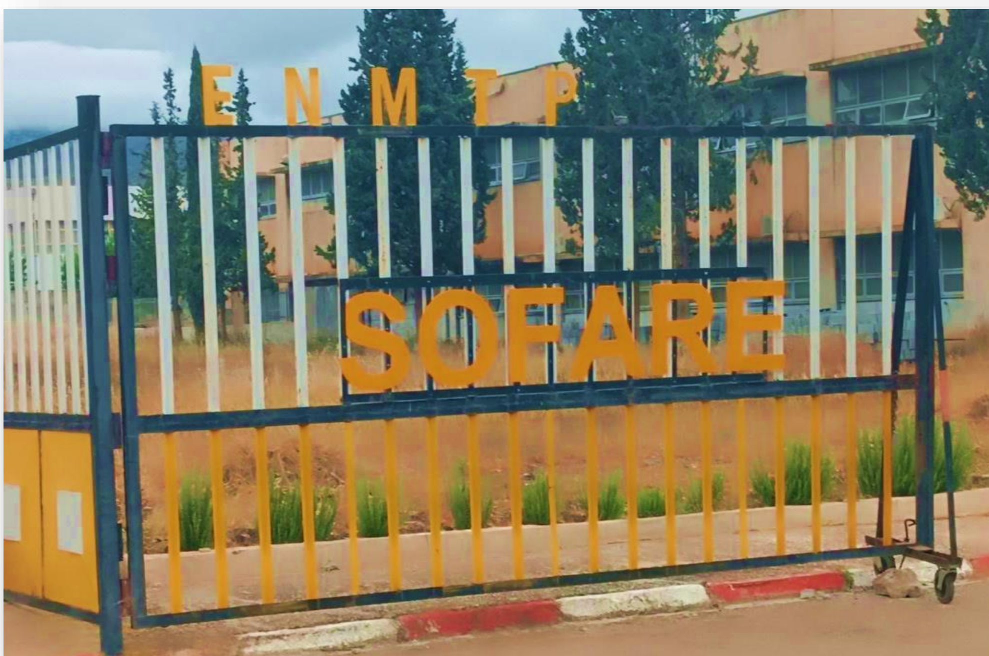

Fig. I.3. The CCA complex

I.6. Organizational chart of the company

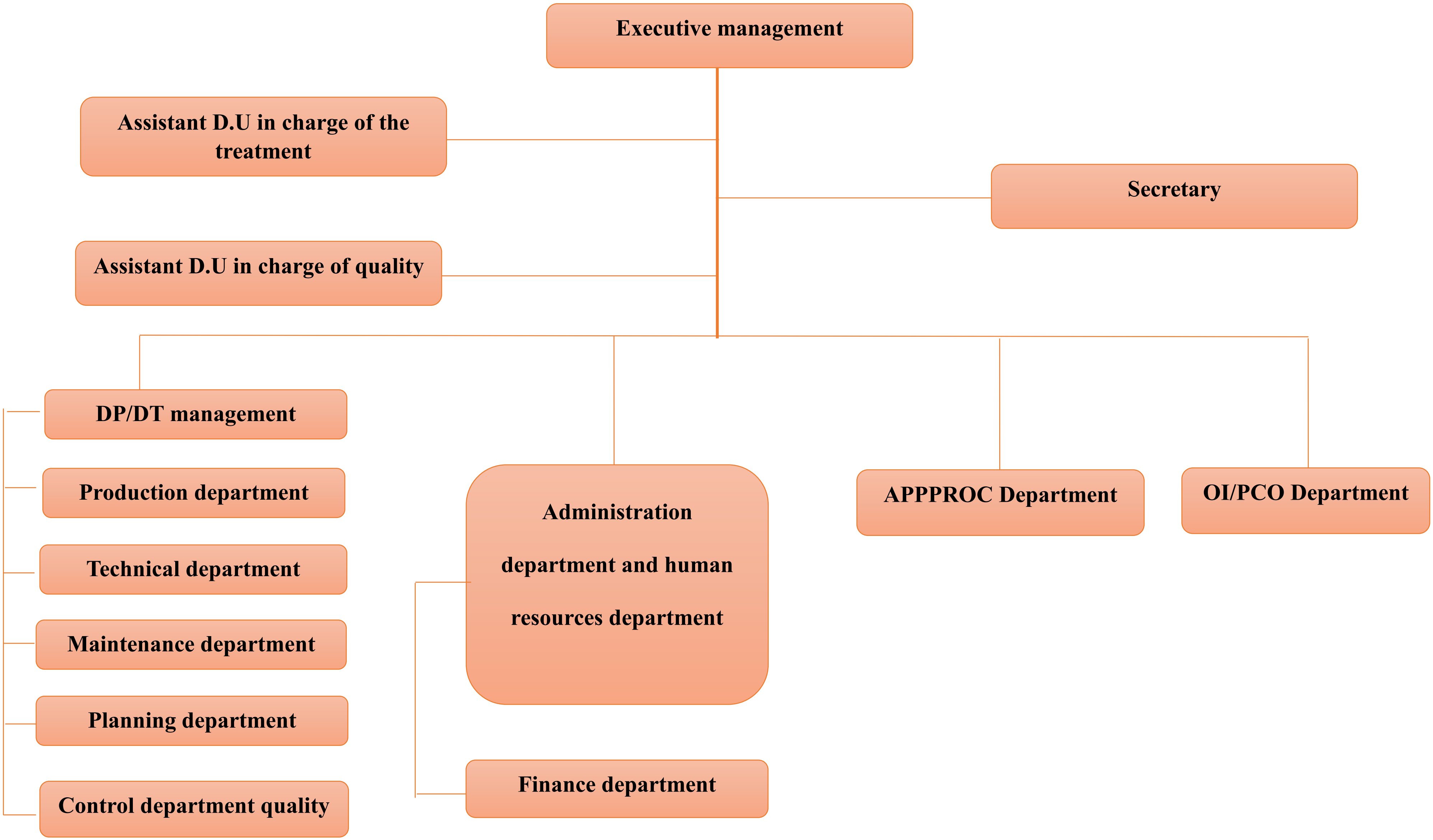

Table I. Company organization chart.

I.7. Activities of SOFARE

SOFARE, with a social capital of 2,000,000,000 Algerian dinars, is engaged in the manufacturing of Backhoe Loaders, Compaction Equipment, Compressed Air Production Equipment, and Concrete Injection Pumps.

Table I.1. Activities of SOFARE

Backhoe Loaders: - Two (02) types of Backhoe Loaders: 4120 S and 4130. - One (01) type of 4×2 Backhoe Loader: 4140 B. - One (01) type of 4×4 Backhoe Loader: 4150.	
Compaction Equipment: - One (01) type of Vibrating Plates: BP 18. - Two (02) types of Single Drum Rollers: SP 24.	
Compressed Air Production Equipment: - Four (04) types of Mobile Compressors: D 45, D 71, and D 103. - Three (03) types of Fixed, Electric Compressors: E 28 and CE 9.	

Concrete Injection Pump (CIP): - One (01) type of Concrete Injection Pump: MI 25.	

I.8. Manufacturing Processes

The manufacturing processes encompass all operations required to construct a technical object. Once the construction requirements are determined in the specifications, the dimensions and shapes of the parts are drawn in the diagrams, the materials to be used are selected, and the procedures are developed in the manufacturing range. The actual fabrication can proceed. This involves several manipulations grouped into three main stages:

1. **Measuring and Marking**: This stage involves measuring the necessary dimensions and marking the outlines of the parts on the raw material. This ensures precision and compliance with design plans.
2. **Machining (or shaping)**: Machining involves transforming raw material into finished parts. This can include processes such as turning, milling, drilling, and grinding. These techniques remove material to achieve the desired shape.
3. **Assembly and Finishing**: Once all parts are manufactured, they are assembled to form the final object. Assembly techniques may include welding, riveting, gluing, and bolting. Finally, finishing touches like polishing, painting, or coating are applied to enhance the appearance and durability of the object.

These stages are critical in ensuring that the final product meets the required specifications and quality standards.[4][5].

At SOFARE, there are many processes:

1. **Cutting and Machining Section**

Table I.2. Cutting and Machining

The processes	Work	The figure
Filming	This is one of the most important manufacturing methods involving material removal on cylindrical parts through which multiple operations are performed.	
Drilling	This is a process aimed at completing holes that contribute to installation.	
Milling	This is one of the most important manufacturing methods involving material removal on prismatic parts through which multiple operations are performed.	

Bore	This is a manufacturing process that involves refining the interior of the cylinder to improve roughness and achieve the highest possible precision.	
Folding **And** **Crunching**	➢ - This is a cold forming process for flat sheets through permanent deformation to bend them at a certain angle as required. ➢ - It is represented by successive folds with dimensions to form a large angle, creating a wide curve through the bending method.	

Chamfer	This is the operation of creating a flattened surface at the end of the tube. The opening created by the chamfering operation allows for welding.	

2. Construction Section:

Table I.3. Construction

Oxy cutting	The basic principle of oxy-fuel cutting was provided by Lavoisier when he conducted the experiment of burning a wire of iron in an oxygen atmosphere.	

Welding	It's a preliminary process that starts with a piece and ends with initial dynamics of forming the final product.	

CNC machine	In the field of mechanical manufacturing, the term "control" refers to any hardware and software whose function is to provide motion instructions to all elements of a machine tool, and at SOFAR, there is a numerical control machine that facilitates a lot of work.	

Chapter II:

About the Back loader

II. About the back loader

II.1. Definition

A backhoe loader is a piece of civil engineering equipment that combines a wheeled loader and an excavator. The excavator, of small size, is mainly intended for light work (such as digging trenches or in agriculture for manure collection). This machine is also called a "loader-backhoe," "digger-loader," or "backhoe." In Quebec, the nickname "pépin" is very common. In French-speaking Switzerland, it is commonly called "trax."

II.2. Scope of application

The backhoe loader can be used for various tasks, such as earthmoving, and offers the advantage of being able to move from one site to another without the need for additional transportation equipment like a low loader or trailer. Its various attachments and accessories, such as the hydraulic breaker, planer, sweeper, as well as the handling hook and snow plow blade, also enhance its versatility. Due to its great versatility, it is difficult to precisely list all the fields of application of the backhoe loader here.

Thanks to its loader bucket at the front and its excavator at the rear, it can perform various operations and is therefore indispensable on multifunctional job sites. With its front loader bucket, it can move and load all kinds of materials, while with its rear excavator and articulated arm, it can penetrate the ground, dig, and clear debris. The loader is present on all earthmoving and construction sites but is also widely used in agriculture. Many attachments and accessories are also compatible with the front loader, adding even more quality and versatility. At the rear, it can be equipped with a hydraulic hammer that allows it to break through obstacles or terrain like a jackhammer, an auger for drilling, a ripper, or grapples.

At the front, it is possible to attach pallet forks, a snow plow blade, or various types of sweepers to collect debris. Considering its ability to perform numerous tasks, it is easier to understand why the loader is a flagship machine in the construction sector. Efficient, high-performing, multitasking, powerful, this machine is a true reservoir of productivity, an indispensable all-in-one tool [6].

II.3. Description:

A backhoe loader consists of 5 essential parts:

- Cabin access door.
- Front bucket.
- Arm with tool (here, a grapple bucket).
- Stabilizer legs.
- Cabin.

II.4. Back loader components:

The backhoe loader, formerly known as the loader-backhoe, has the advantage of being a wheeled vehicle. Therefore, it can be driven on public roads and does not necessarily need to be transported on a low loader or trailer. Its maximum speed, between 40 and 45 km/h, allows it to move around job sites as long as they are not too far from the warehouse.

There are three parts to a backhoe loader: the front part (loader), the cabin, and the rear part (excavator). The loader bucket can be raised and lowered using a lifting arm, while the excavator's articulation involves a telescopic arm that allows the backhoe bucket to rotate 90°. All of this is powered by hydraulic cylinders. A diesel engine provides all the necessary power for this equipment.

There are backhoe loaders of different power levels; the greater the power, the easier it is to overcome obstacles. The same applies to the driving wheels: a machine with 4-wheel drive will move more easily on rough terrain. The following figure illustrates the essential parts of the backhoe loader [6].

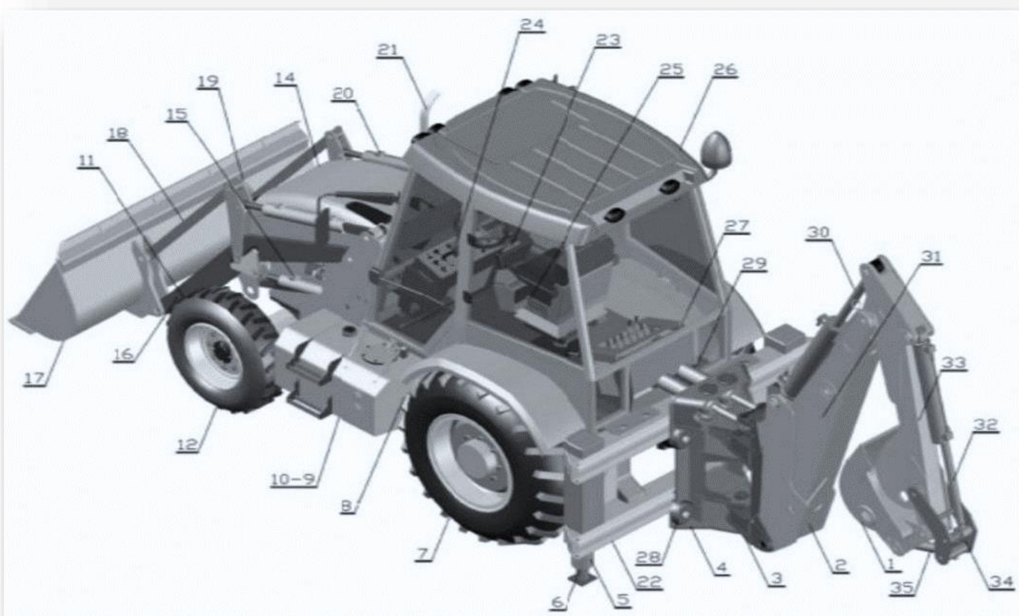

Fig. II.1. Different parts of the backhoe loader.

Table II.1. Different parts of the backhoe loader

1	Backhoe Bucket (Backhoe Bucket): This is a component of the excavator that the operator loads by making a movement closer to the tractor, allowing for digging into the ground. The teeth of the bucket used have a significant influence on the efficiency of the work performed.
2	Boom: This is a component whose head moves up and down, allowing for the raising or lowering of the backhoe bucket.
3	Swing Column: This is a component that allows the backhoe arm and bucket to rotate horizontally, providing the machine with a wide range of motion for excavation and loading tasks.
4	Sliding Support: This component provides stability and support to the backhoe arm, allowing it to extend and retract smoothly during operation.
5	Stabilizer Cylinder: This component is responsible for extending and retracting the stabilizer legs of the backhoe loader, providing stability to the machine during operation.
6	Stabilizer Foot: This is the part of the stabilizer system that makes contact with the ground to provide additional stability to the backhoe loader during operation.
7	Rear Wheel: This is the wheel located at the back of the backhoe loader, which provides traction and supports the machine's movement on the ground.
8	Gearbox: This is the component responsible for transmitting power from the engine to the wheels of the backhoe loader, allowing the operator to control the speed and direction of the machine.
9	Fuel Tank (on the right): This is the container where fuel is stored for the engine of the backhoe loader, located on the right side of the machine.
10	Hydraulic Tank (on the left): This is the container where hydraulic fluid is stored for powering the hydraulic systems of the backhoe loader, located on the left side of the machine.
11	Front Axle: This is the component that connects the front wheels of the backhoe loader, providing support and allowing for steering control.
12	Front Wheel: This is the wheel located at the front of the backhoe loader, which provides traction and supports the machine's movement on the ground.
13	Hydraulic Pump: This is the component responsible for generating hydraulic pressure in the hydraulic system of the backhoe loader, powering the various hydraulic functions such as lifting, digging, and moving attachments.
14	Bonnet: This is the protective cover that encloses the engine compartment of the backhoe loader, providing access for maintenance and protecting the engine from external elements.
15	Lifting Cylinder: This hydraulic device on the front loader consists of telescopic arms that enable the operation of the articulated lever, allowing for lifting and lowering of the loader bucket.

16	Lifting Arms: These are the two levers that connect the loader to the tractor and enable it to be raised or lowered.
17	Loader Bucket 4-in-1: Designed for various applications in construction, landscaping, demolition, and recycling, the "4-in-1" bucket is a versatile tool that performs the roles of multiple attachments in one. Used like a regular bucket, the "4-in-1" bucket enables digging, leveling, pushing, transporting, loading, and unloading of materials.

18	Rollover Bar: This is a safety feature typically found on heavy machinery like tractors and backhoe loaders. It's designed to protect the operator in the event of a rollover by providing a protective structure around the operator's compartment.
19	Pantograph: This is a mechanism often used in backhoe loaders for extending and retracting the digging arm. It consists of multiple linked sections that can adjust their length to reach varying depths during excavation.
20	Tilt Cylinder: This hydraulic component is responsible for controlling the tilting motion of the backhoe loader's attachment, such as the loader bucket or the excavator arm, allowing for precise positioning and manipulation of materials during operation.
21	Diesel Engine: An internal combustion engine in which air is compressed within the cylinders. This compression raises the air temperature to a point where it ignites the injected fuel towards the end of the compression stroke.
22	Chassis: The chassis is the structural framework of the backhoe loader that supports the various components such as the engine, hydraulic systems, operator cab, and attachments. It provides stability and durability to the machine.
23	Loader Valve: This is the hydraulic valve responsible for controlling the operation of the loader attachment on the backhoe loader. It regulates the flow and direction of hydraulic fluid to control the movement of the loader arms, bucket, and other loader functions.
24	Steering Box: This is the component responsible for translating the driver's input from the steering wheel into movement of the front wheels, allowing the operator to control the direction of the backhoe loader.
25	Driver's Cab: This is the enclosed compartment where the operator sits to control and operate the backhoe loader. It typically contains the steering wheel, controls for various functions, instrumentation, and seating for the operator.
26	Cab: This is the enclosed space from which the operator maneuvers the backhoe loader using the various control levers at their disposal.
27	Excavator Valve: This is the hydraulic valve responsible for controlling the operation of the excavator attachment on the backhoe loader. It regulates the flow and direction of hydraulic fluid to control the movement of the excavator arm, bucket, and other excavator functions.
28	Locking Cylinder: This hydraulic component is responsible for locking or securing specific parts of the backhoe loader in place, such as stabilizer legs or attachment arms, to ensure stability and safety during operation.

29	Rotation Cylinder: This hydraulic component is responsible for controlling the rotational movement of the backhoe loader's attachment, such as the excavator arm or the loader bucket. It allows for precise rotation and positioning of the attachment during operation.
30	Excavation Cylinder: This hydraulic device, with the help of telescopic arms, enables the movement of the main arm of the excavator.
31	Boom Cylinder: This hydraulic device, with the assistance of telescopic arms, facilitates the movement of the boom on the backhoe loader.
32	Bucket Cylinder (Backhoe Bucket Cylinder): This hydraulic device, with the assistance of telescopic arms, controls the back-and-forth movement of the backhoe bucket.
33	Excavator Arm: This is the part of the backhoe loader that allows the excavator to perform a swinging motion to move the bucket closer to or further away from the tractor.
34	The pantograph is a mechanical or hydraulic device used in certain types of mechanical shovels or excavators. It allows for adjusting the horizontal reach of the working tool, such as the bucket or grapple, by moving it forward or backward relative to the base machine. This provides greater flexibility and precision during excavation or material handling.
35	The rollover bar is a safety feature typically found on heavy machinery such as tractors or backhoe loaders. It's a sturdy bar or frame structure installed above the operator's compartment. Its purpose is to protect the operator in the event of a rollover or overturn of the machine. If the machine were to tip over, the rollover bar prevents the operator from being crushed by the weight of the equipment. Instead, it provides a protective space around the operator, reducing the risk of serious injury or fatality.

II.5. Technical characteristics of the back loader:

The retro loader 4150 4x4 has the technical specifica presented in the tables, which meet the requirements of the customers (See Figure III.2).

Fig. II.2. The 4150 4x4 backhoe loader

- **technical characteristics of the engine**

Table II.2. Engine characteristics

Manufacturer	DEUTZ-SAFMMA
Type	BF4M2012
Cooling	Water
Maximum power at 2500tr/min	65kw/88cv
Maximum torque at 1500tr/mn	335 N.m
Electric start	12v / 4kw
Suction	Supercharged
Fuel consumption	220g/kwh

- **Technical specifications of electrical equipment**

Table II.3. Electrical equipment

Alternator	12V55A
Battery capacity	110 Ah

- **Technical specifications of transmissions**

Table II.4. Transmissions

Power Shunt gearbox with torque converter	
Movement speed (kW/h)	1st :6.6 2nd :10 3rd:23 4^{th} :40
Forward	1st :6.6 2nd :10 3rd :23 4th :40
Order	Mechanical

- **Technical specifications of axles**

Table II.5. Bridges

Allowable (dynamic) load capacity of the front axle	9500kg
Allowable (dynamic) load capacity of the rear axle	7500kg

- **Technical specifications of braking system**

Table II.6. Braking

Service brake	Hydraulic multi-disc brakes with integrated oil bath at the rear axle
Parking brake	Disc brakes integrated into the rear axle, with mechanical control

- **Technical specifications of the drivetrain**

Table II.7. Driving

Direction	Hydrostatic type
Outer turning radius in forward motion	5.5m
Steering angle	+/-25°

- **Technical specifications of the reservoir**

Table II.8. Reservoir

Fuel tank capacity	120L
Maximum visual level capacity of the hydraulic reservoir	100L

II.6. Equipment and accessories

Front-end loaders are machines that can be equipped with numerous different equipment or accessories, making these machines particularly versatile.

Loader side:

The 4x4 loader is composed of a loader arm, linkages, and a bucket. The loader fills by moving the machine forward, with a bucket designed for various applications in construction, landscaping, demolition, recycling, and material transport. The "4-in-1" bucket is a versatile tool that serves multiple accessory roles in one.

When used as a bucket, the loader can load, level, push, move, and dump material. However, the container holding section can open and close on the material because it has a fixed set. This feature is particularly useful for cleaning work. Additionally, the grapple bucket can be opened to release material without the need to tilt the bucket forward, which is advantageous when unloading into high-sided trucks.

Finally, the grapple bucket also allows the "4-in-1" bucket to be used as a grabber to pick up pieces of concrete, brush, or other difficult-to-handle materials.

II.7. Identification of floorboards and loader manipulators

✓ Manipulators

The technique employed in the 4TH6N and 4TH5 manipulators differs from that of the 4TH6. It provides a reduction in variations of the perceived effort during the tilting of the control lever. The 4TH5, being smaller and lighter, is particularly intended for use on compact machines.

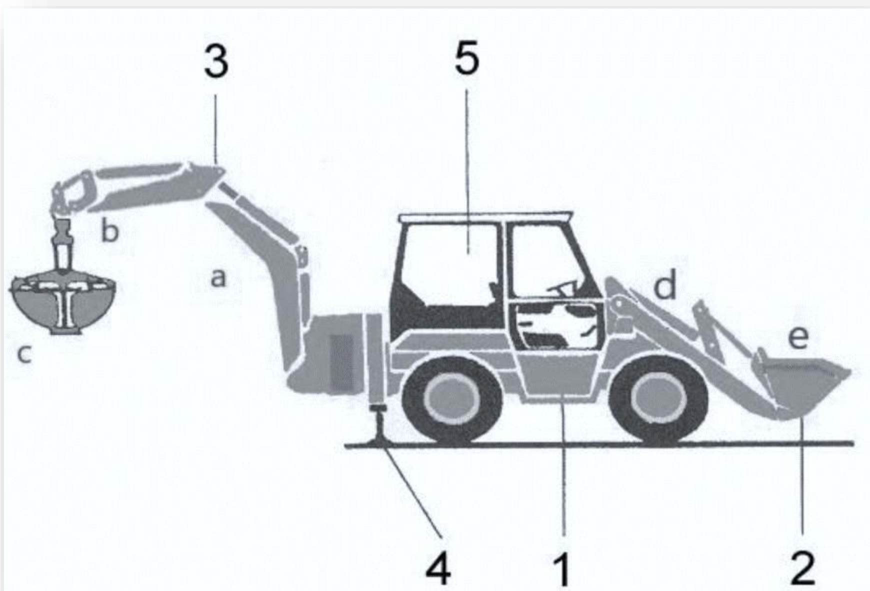

Fig. II.3. retro charger.

a: Main arm.	1. Access port to the cabin.
b: Excavation arm.	2. Loader bucket.
c: Back bucket.	3. Boom with tool.
d: H frame.	4. Stabilizing outriggers.
e: Front bucket.	5. Cabin with driver's seat.

Chapter III:

Exploring Godet and Incident Analysis

III. Exploring Godet and Incident Analysis

III.1. Exploring Godet

III.1.1. Introduction

We will begin by providing a more detailed explanation of the Godet, followed by an exploration of the materials used in its manufacturing process. Subsequently, we will delve into the welding procedure applied to it. Finally, we will elaborate on the issue that occurring in detail.

III.1.2. Definition of Bucket

A "bucket" is typically defined as a round container with a flat bottom and a curved handle, used for holding, carrying, or storing liquids and solids. It can be made from various materials such as metal, plastic, or wood. In addition to its primary function, "bucket" can also refer to specific components in machinery, such as the scoops on excavating machines or the receptacles on waterwheels [3].

Fig. III.1. Godet 4041 4*4.

III.1.3. The materials

The materials used in making Godet are as represented in the following table

A table representing the materials used in making Godet

Table III.1. The materials

Rep	**Designation**	**Matière**
01	Axe	42Cr.Mo4
02	Chape	ST 52-3

- 42CrMo4 is a type of alloy steel known for its excellent combination of high strength, toughness, and wear resistance. It is often used in applications requiring high tensile strength and good impact resistance, such as in the manufacturing of gears, shafts, and other machine components subjected to heavy loads and stresses. Additionally, 42CrMo4 offers good hardenability and can be heat treated to achieve desired mechanical properties, making it versatile for various engineering applications.

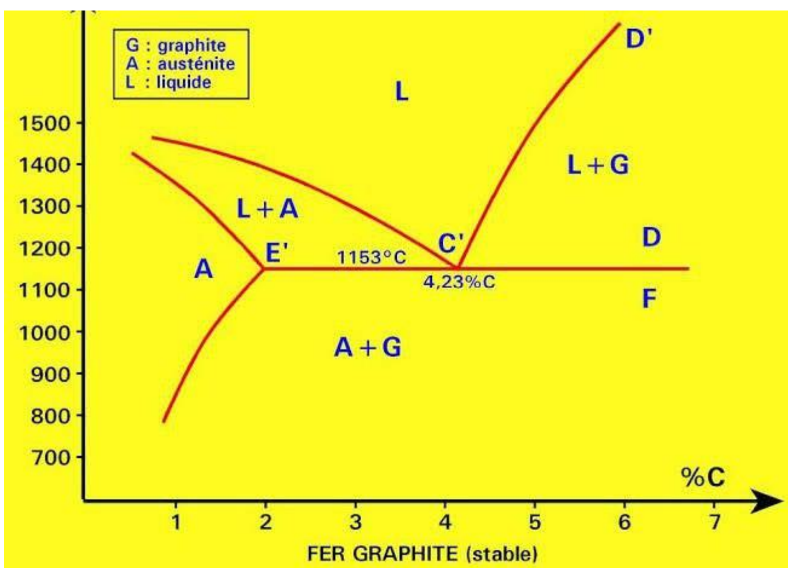

Fig. III.2. Cast irons in the iron-carbon diagram

- ST 52-3 is a type of structural steel commonly used in various engineering applications. This steel is known for its good resistance to dynamic impacts and corrosion, making it ideal for use in manufacturing metal structures such as solid frames, pipes, and bridges. ST 52-3 also exhibits high tensile and compressive strength, making it a common choice in industries like shipbuilding, railways, automotive, and heavy equipment.

III.1.4. Welding Process

MAG Welding

❖ **Consumable Products:**

• Wire Electrode: Continuously fed wire acting as the electrode.

• Gas Shielding: Inert gas (MIG) or active gas (MAG) used to protect the weld from oxidation.

❖ **Operating Modes:**

• Parameter Adjustment: Voltage, current, gas flow rate, wire feed speed.

• Edge Preparation: The edges of the pieces to be welded must be clean.

• Welding Position: Positions based on the joint type and process (MIG or MAG).

• Control of Torch Angle: The angle between the torch and the workpiece.

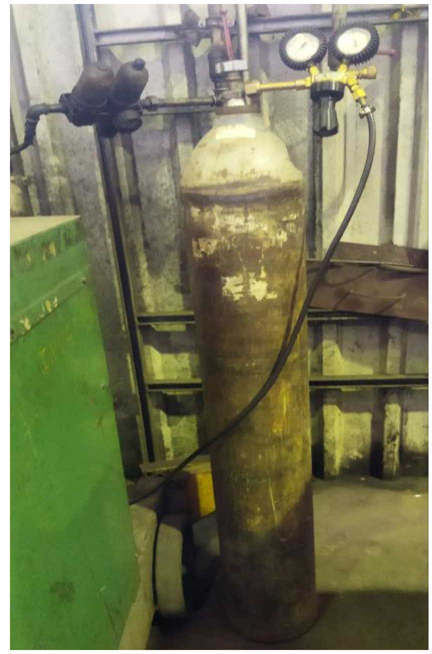

Fig. III.3. A mixture of argon and Co2

III.1.5. Welding Control

III.1.5.1. Explain the process (controlled Deposition Method CDM)

Stages of quality control in welding typically involve various procedures and techniques, including:

- **Parameter Adjustment:** This involves adjusting the current, voltage, and wire feed speed to ensure appropriate welding conditions.
- **Raw Material Inspection:** Inspection of raw materials such as wires and gases to ensure their quality and compliance with standards.
- **Surface Preparation:** Cleaning and preparing the surface of the workpieces before welding to ensure continuity of the weld and quality of the joint.
- **Non-Destructive Testing (NDT):** Using techniques such as X-ray inspection or ultrasonic testing to detect defects without damaging the workpiece.
- **Destructive Testing:** These tests include techniques like bend tests and fracture tests to verify the strength of the weld.

The **"C**ontrolled **D**eposition **M**ethod**"** is considered part of these processes where material deposition is precisely controlled to provide and direct metal in an accurate and controlled manner. This helps in reducing defects and improving the quality of the final joint.

The process of leak detection conducted after the welding process using white and red materials is typically classified under **N**on-**D**estructive **T**esting **(NDT)** procedures. This process is used to detect any leaks or defects in the weld joint after the welding process is completed.

White and red materials are used to detect leaks or defects in the weld by applying them to the weld surface after it has cooled. Usually, the white material is a powder applied to the weld surface and is used to detect surface leaks, while the red material is a liquid applied to the weld and is used to detect deeper leaks.

The leak detection process using white and red materials is an important part of the welding quality verification stages, as it helps identify any potential leaks or defects beneath the weld surface that may affect the quality of the final joint.

III.1.5.2. Conduct an experiment on the topic

In these following pictures, we conducted a small experiment with the team at Sofar Company regarding this method for monitoring gaps and leaks that occur after welding, and this is what the following pictures illustrate.

III.1.5.2.1. Preparation of materials

We are preparing the red and white bottles for later use in the leak detection process, along with two metal plates for the experiment:

Fig. III.4. White Contrast Liquid and Red Contrast Liquid

Fig. III.5. Two metal plates for the experiment

III.1.5.2.2. Clean the two metal pieces

After preparing the necessary materials, we remove rust from the area to be welded by scrubbing it with a specialized tool, and then using the welding cleaning bottle to remove all unnecessary residues in preparation for the welding process.

Fig. III.6. Two metal plates for the experiment

Fig. III.7. A special bottle for cleaning parts before welding

III.1.5.2.3. Welding the two pieces

We are now welding the two pieces together. Here, the team intentionally leaves gaps so that we can observe them during the welding quality test and see how they appear.

Fig. III.8. Welding the two pieces

III.1.5.2.4. Paint with white liquid

And here, after finishing the welding, we leave the two metal pieces for a certain period until they cool down, then we coat them with the white liquid as shown in the following picture:

Fig. III.9. Paint with white liquid

III.1.5.2.5. Paint with red liquid

After painting them white, we wait for a few minutes until it dries, then we proceed to paint the second face of the metal in the welding area:

Fig. III.10. Paint with red liquid

III.1.5.2.6. Leakage checking

We return to the white-painted surface and then wait for a few minutes until the process activates. Afterwards, red dots appear on the white-coated surface, indicating that those areas have leaks, and we need to re-weld them. This is an example illustrated in the following picture:

Fig. III.11. Appearance of red dots on the surface

III.2. Incident Analysis of Godet

III.2.1. Clarifying Current Impact:

The carefully examined problem causes numerous issues, such as material losses like having to purchase a new product, wasting time and money on repairs, and re-manufacturing the broken part. Consequently, the company faces a breakdown and enters a dangerous slope of losses and debts, necessitating a thorough study of the problem and its resolution.

III.2.2. Study and analyze the problem:

III.2.2.1. The Problem zone.

The following image illustrates the area that is continuously subjected to breakage, highlighted by the yellow circle.

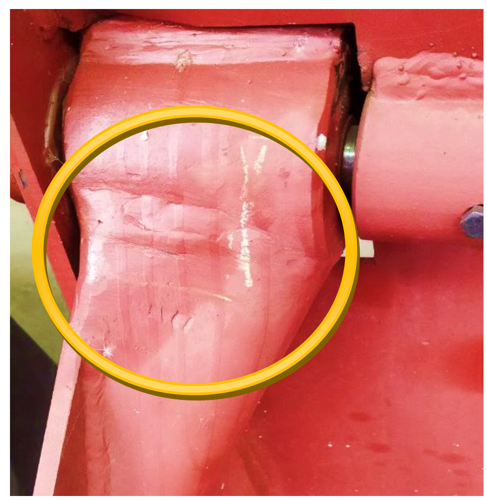

Fig. III.12. A picture showing the fracture areas in the Godet

Fig. III.13. A closer picture of the fracture area

III.2.2.2. Problem analysis

After conducting specialized research on the studied area, we found that the occurrence rate of this problem reaches 73%. We observe each time the breakage occurs in the welding area surrounding the two pieces, sometimes followed by cracks, some of which are deep and others light.

This happens when the weight borne by the excavator reaches a certain limit, up to 222 kg, which occurs from its first use.

The incidence of the problem	**Types of cracks**	**Location of fracture**	**The time when the breakage occurs**
The incidence of the problem is 73%	Deep cracks and medium cracks	Welding line	Once the load weight of the excavator bucket reaches 222 kg

Table III.2. Problems analysis table

Chapter IV:

Study the problem and solve it

IV. Study the problem and solve it

IV.1. Study the problem

IV.1.1. Introduction

In this section, we will delve into studying suitable solutions to address the previous issue by following crucial steps outlined in a structured plan.

This plan involves proposing hypotheses about the root cause of the problem, followed by identifying the correct hypothesis through thorough examination and analysis. Subsequently, we will devise scientific solutions that are apt for resolving the issue.

Finally, we will select the most suitable solution. We will then conclude by reviewing the preceding steps and seeking feedback from engineers and supervisors regarding the proposed solution.

IV.1.2. Proposing hypotheses

To find a solution to this problem, we choose to use the Ishikawa method, which leads us to all possible causes of this problem, represented in the following diagram.

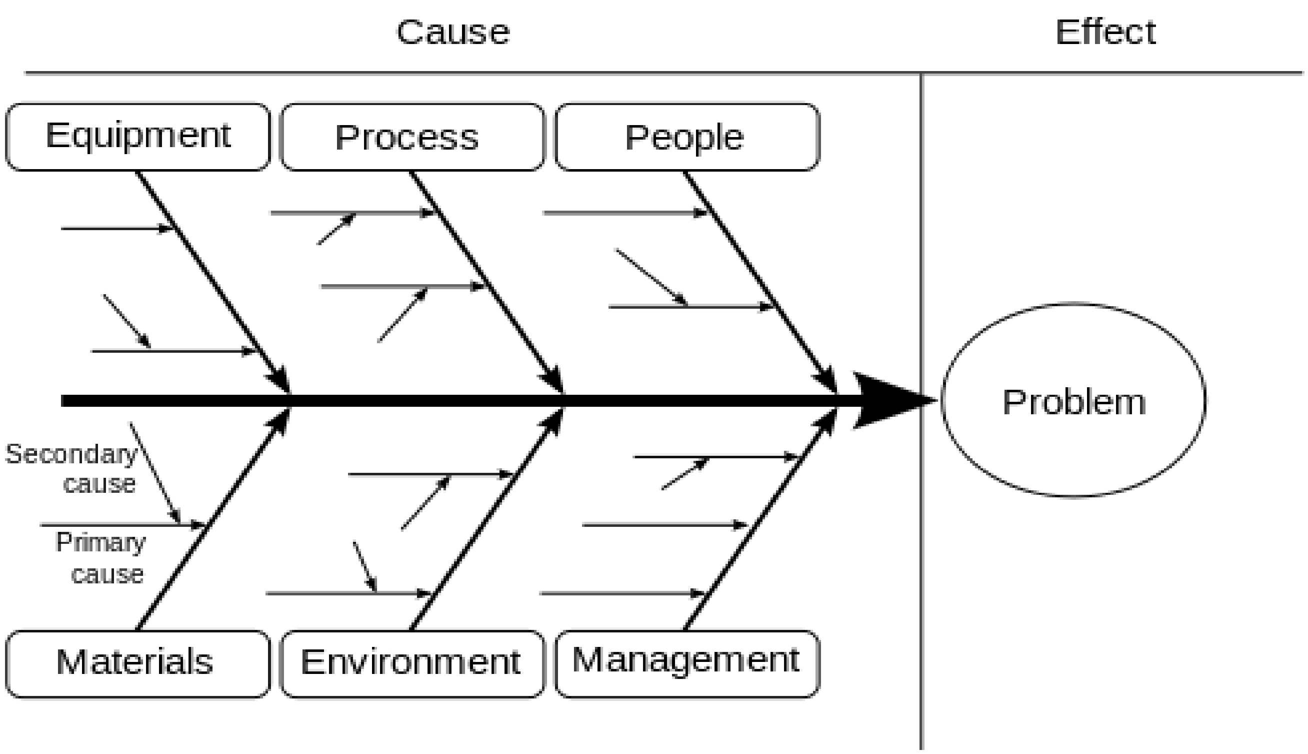

Fig. VI.1. Diagram showing the Ishikawa method

By following the Ishikawa pre-planned approach, we gather all possible causes of this problem in the following table.

Table.VI.1. Table showing possible causes of Ishikawa

Equipment	Process	People	Materials	Environment	Management
1-The mag could be spoiled.	**1-**The cleaning method for the piece could be incorrect. **2-**The welding angle may be incorrect. **3-**The welding may not be sufficient to secure the piece. **4-**The design may not be optimal.	**1-**The workers may be fatigued. **2-**The workers may lack training. **3-** Labor shortage. **4-** Continuous absence of supervisors.	**1-**The material used for welding may not be suitable. **2-** The main material may not be suitable. **3-** The welded metals may not be compatible with each other.	**1-** Moisture levels could be high. **2-** The temperature may not be suitable. **3-** A polluted environment.	**1-** Lack of worker supervision. **2-** Non-compliance with quality and manufacturing regulations. **3-** Negligence in repairing machinery and equipment.

IV.1.3. Selecting the most appropriate hypothesis and proving it

Through our study of each individual cause, we concluded that the primary reason lies in the design of the welded piece, making it difficult and insufficient to weld properly, thus resulting in the instability of the welded piece, then the reason lies in **Process** Specifically, in **(**The welding may not be sufficient to secure the piece**)** and **(**The design may not be optimal**)**.

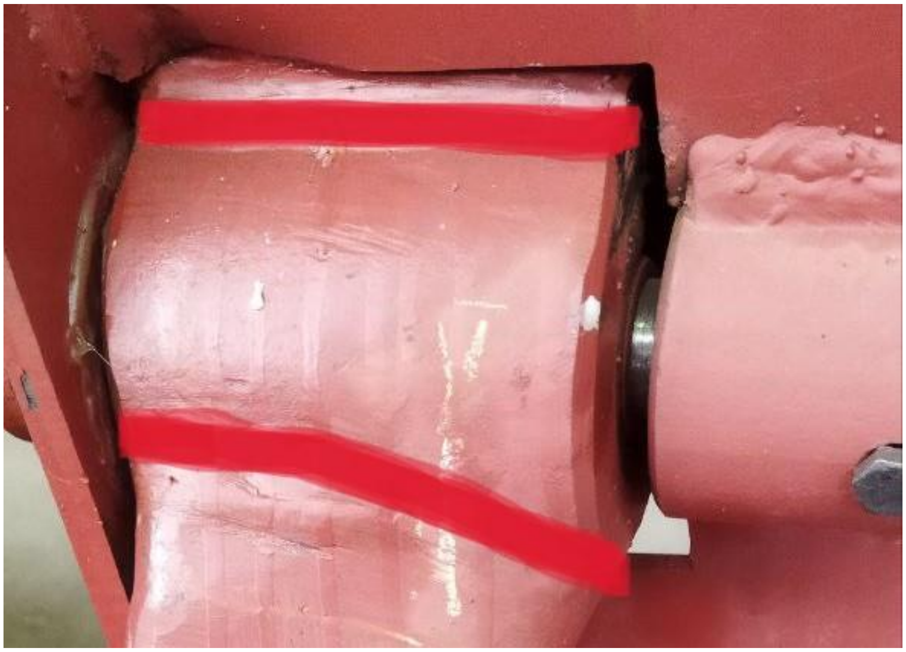

Fig. IV.2. A picture showing the welding lines

IV.2. Solving the problem

IV.2.1. Searching for possible solutions

The following section presents possible solutions:

1. **Welding a support piece**

We add a support piece between the two parts that frequently experience fractures by welding it above them. This is to increase the load-bearing capacity and withstand the pressures caused by the excessive weight of the drilling machine, thus avoiding its breakage, As illustrated in the following image:

Fig. IV.3. An image showing how to weld a supporting piece

2. Reducing the necessary load for the Godet

We avoid carrying weights exceeding the capacity of the Godet without altering its design, meaning we use it to lift weights not exceeding 222 kg to avoid breakage

3. Changing the design of the Godet

We change the design of the crane to provide a more suitable space and angle for welding the two pieces together perfectly. This will result in greater load-bearing capacity and prevent breakage

4. Changing the materials

Changing the materials used in manufacturing the excavator to new materials with greater capacity for bearing heavy loads

IV.2.2. Choosing the most suitable solution and studying it

IV.2.2.1. Analysis of the solutions

After presenting the possible solutions earlier, we will choose the appropriate solution from among them and study it further:

- **Welding a support piece**

This solution is highly effective, but it may not be suitable for a large company like **Sofar**. However, it is very useful for temporary use, such as in emergency situations.

- **Reducing the necessary load for the Godet**

This solution is simple but not suitable for heavy loads, making it an unsuitable and unacceptable solution for a strong brand like **Sofar**'s drilling machine.

- **Changing the design of the Godet**

This solution may be considered ideal, excellent, and highly effective as it addresses and covers all previous problems, and we do not see any drawbacks in it.

- **Changing the materials**

The metals used in drilling machine manufacturing are carefully selected, so we do not find any metals better than the ones currently used. Therefore, this solution is not suitable for solving the current problem.

IV.2.2.2. The final solution

Through our previous study and analysis of the solutions, we have concluded that the optimal and best solution is to create a new design for the constantly breaking part **(Changing the design of the Godet)**. This is to provide sufficient space for welding and securing it fully and permanently.

IV.3. Implementing the researched solution

The main objective is to design a new shape that allows us to weld the two pieces together in a way that makes them stronger and more resilient to pressures:

IV.3.1. Designing the new shape

First, we design the new structure that allows us to create a longer welding line, as shown in the following images, this is done by using the famous SolidWorks software to design and assemble the pieces and structures as an initial shape:

IV.3.1.1. Definition of SolidWorks

SolidWorks is a popular CAD/CAM software used for designing and assembling three-dimensional mechanical parts. It provides an easy-to-use interface with a wide range of tools for creating and editing 3D models.

The software is widely used in various industries such as mechanical engineering, automotive, aerospace, marine industries, and many others.

Users can use SolidWorks to create precise drawings, perform engineering analysis, create complex assemblies, generate designs for 3D printing, and much more.

Figure VI.4. software logo.

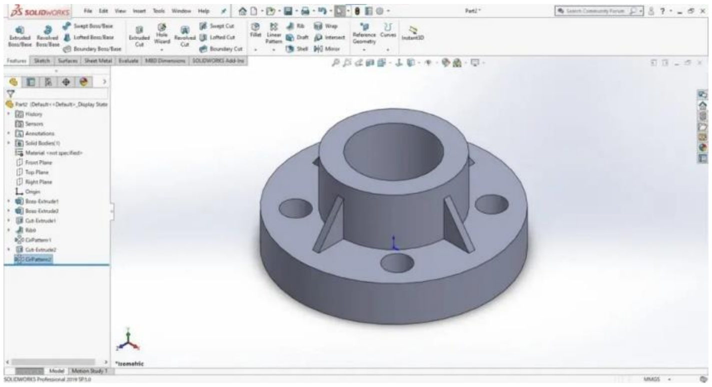

Figure VI.5. Software interface

IV.3.1.2. Implementing the design

After familiarizing ourselves with SolidWorks and after analyzing and discussing the appropriate shape that will solve the problem, we concluded that the optimal and best shape is represented by the following images:

IV.3.1.2.1. Structural design

First, we design the structure of the crane without the piece we want to change, using SolidWorks software:

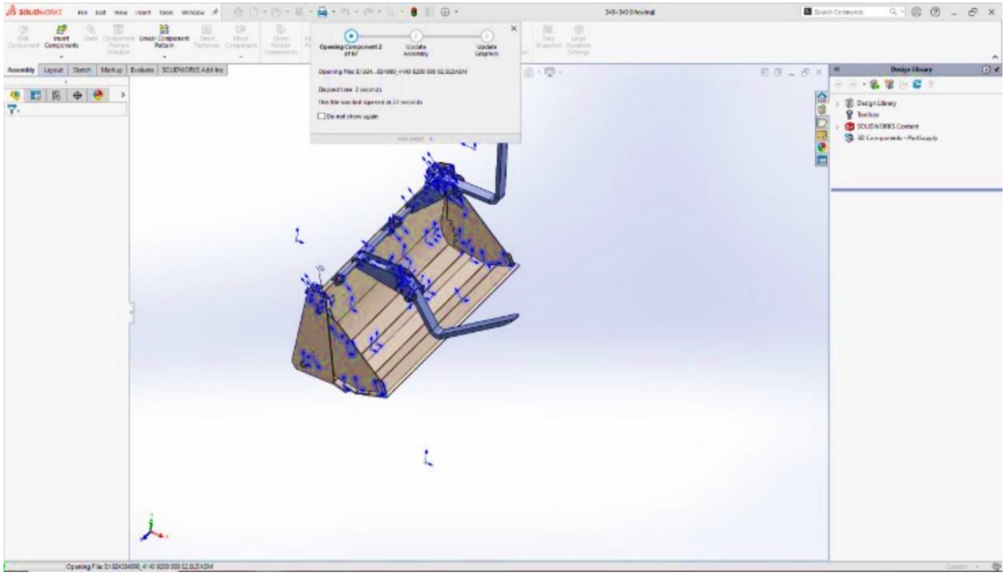

Fig. VI.6. Structural design

IV.3.1.2.2. The new piece designs

Secondly, we design the new piece on which our study is based, by providing a suitable line for the long welding:

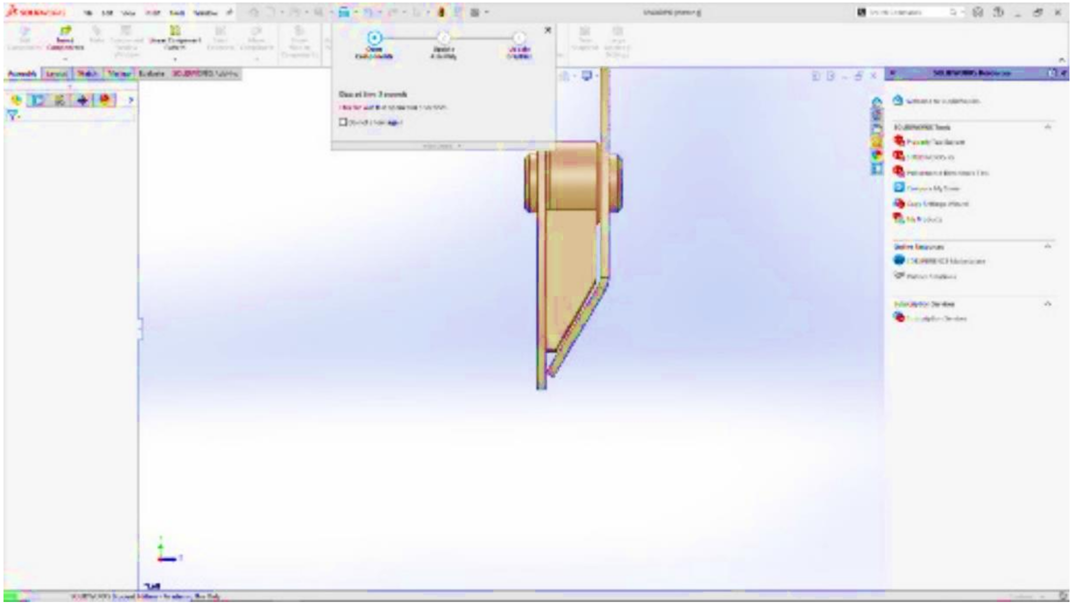

Fig. VI.7. The new piece designs

IV.3.1.2.3. Assemble the piece with the structure

Now, we assemble the design of the new structure with the design of the new piece to obtain the complete design, which will be the new version that solves the problem:

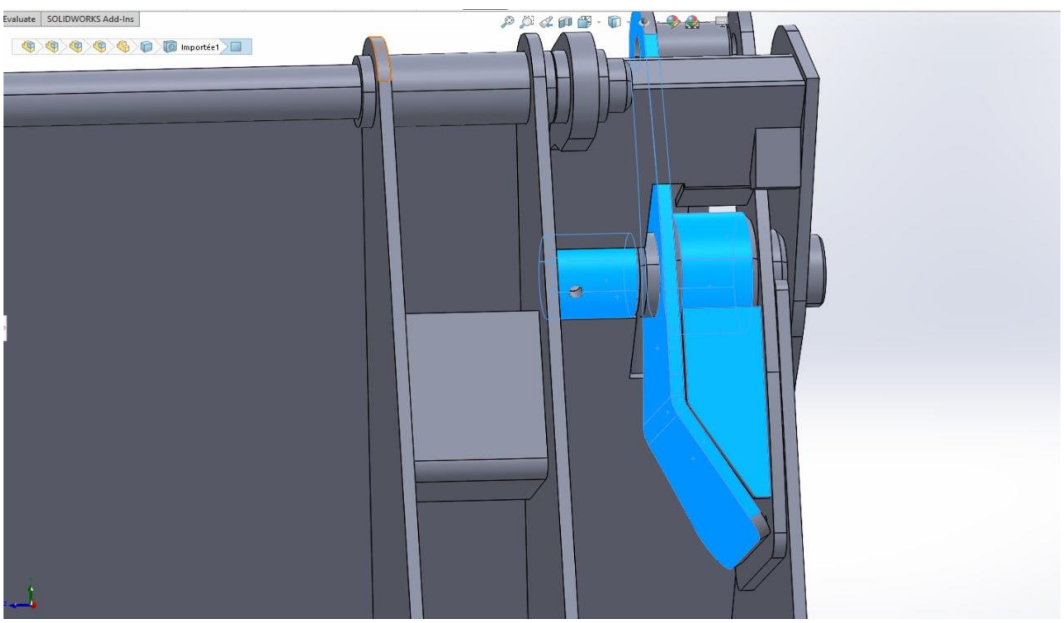

Fig. VI.8. Assemble the piece with the structure

IV.3.2. Implementing the design

After implementing the new design in reality, the result was as shown in the figure below. We notice that the welding line has become longer, allowing us to weld the two pieces together perfectly. The process was successful after testing with heavy weights:

Fig. VI.9. The final shape after implementation in reality

IV.4. Review and opinions

IV.4.1. Review of the preceding

First, we introduced everything related to the **Sofar** company. Then, in the second chapter, we provided a complete explanation of the studied drilling machine. In the third part, we explained the excavator bucket and the problem that occurs with it. In the fourth part,

We used the Ishikawa method to identify all possible causes of this problem and identified the root cause. Then, we proposed a set of solutions and studied each one separately to choose the most suitable and optimal solution. We then drafted it using SolidWorks software and applied it in reality, testing it.

IV.4.2. Opinions of engineers and supervisors

After presenting our study to the supervisors at **Sofar** Company, they were impressed with our analytical approach and problem-solving skills, leading us to this excellent and fantastic solution. To our surprise, **Sofar** Company had reached the same root cause of the problem and chosen the same solution, which is redesigning to ensure a longer welding line.

They had recently started implementing it. This helped us find recent pictures of the solution being applied in reality, which was a source of pride for us and for our prestigious institute. It taught us the methods of analysis and solution-finding through these studies and research conducted together over the years of study.

General Conclusion

General Conclusion

The study conducted on the excavator bucket enabled us to reach a final solution to the problem of weld breakage by designing a new bucket that allows for a longer welding line while ensuring proper performance through experimentation. This provides greater comfort for welders and workers, as well as enhancing **Sofar** Company's reputation and solving the recurring breakage issue.

Reference

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